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Implementation of Low Power Approximate Multipliers using Exact and Inexact Adders in Error Tolerant Applications

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***Abstract*— Various multiplication techniques which find immense benefits in real time applications have been worked on and continues further to design power and area efficient multipliers with increased performance as well. On comparing multipliers designed using exact and inexact adders along with low voltage-low current adders leads to the design of approximate multipliers, that focuses on the idea of trading precision to design power and area efficient circuits. Applications like error tolerance use approximation to tradeoff accuracy and obtain low area/power/delay circuits. The proposed work focuses on designing 4-bit and 8-bit approximate multipliers using inexact adders and Error Prediction and Correction Logic along with low voltage-low power adders such as Ripple Carry Adder (RCA), Carry Save Adder (CSA), Carry Skip Adder (CSKA) and Carry Look-Ahead Adder (CLA) which are beneficial in Error tolerant applications inferred from the Relative Error Distance (RED)/Mean Relative Error Distance (MRED) calculations. The proposed multipliers are simulated using VIVADO 2024.2 design tool and implemented on AMD Zynq 7000 SoC ZC702 (Zed Board) Evaluation Kit FPGA board. It is depicted from the tabulations that approximate multipliers designed using inexact adders is more power and area efficient than approximate multipliers designed using EPCL and exact adders. Hence this multiplier can be used in future work to design higher bit for applications like processing elements (PE) in AI accelerators.**

Keywords*— Approximation, Accuracy, Inexact adders, EPCL, Power, LUTs, RED

I. INTRODUCTION

Unlike the conventional multiplier circuits, the approximate multipliers are designed where slight inaccuracies are acceptable for significant gains in power efficiency, area reduction and higher performance. Approximate Multipliers find application in error-resilient fields like Digital Signal Processing (DSP), Wireless communication, Robotics, Multi-media, and Image processing.

The various strategies implemented in designing approximate multipliers as proposed in work [1-2] include: approximate adders, strategic placement of gates, approximate algorithms, truncated multiplication, and approximate arithmetic techniques. These techniques are applied across various multiplier architectures in [3-4] that includes array multipliers, Booth multipliers and Wallace tree multipliers. This work demonstrates the approximate multiplier circuits design using exact and inexact adders along with various low power-low current consuming adders like Ripple Carry Adder (RCA), Carry Save Adder (CSA), Carry Skip Adder (CSKA) and Carry Look-Ahead Adder (CLA) and Error Prediction Correction Logic (EPCL) with Ripple Carry Adder (RCA), Carry Skip Adder (CSKA) and Carry Look-Ahead Adder (CLA) to minimize power and area.

Design synthesis in work [7-9] has been done using System Verilog many multipliers ranging from 4 to 128 bits. Addition- ally from work [10], with a median inaccuracy of 1.39 to 3.32 percent, these inaccurate multipliers save an average of 31.78 to 45.4 percent in power relation to similar accurate multiplier designs. Furthermore, an 8-bit approximate multiplier can be designed using approximate compressors which enhances the accuracy and improved performance in image processing applications. Also, dividing partial products into Look Up Table (LUT) based implemented parts show improvised results than the conventional multipliers.

II. PROPOSED WORK

A. Exact and Inexact Adders

The exact adders from existing work [7] produce accurate output results when used in designing various other digital circuits. Inexact adders are implemented in multiplier circuits for partial product addition operations by trading off accuracy for enhancing power and area. The figure 1 shows block diagram of 4-bit approximate multiplier circuit using 2-bit multipliers represented by M0, M1, M2 and M3 with 4-bit operands A and B producing partial products i_K, i_L, i_M and i_N of 4-bits each and produces the final output o_Y which is of 8-bits using the exact and inexact adders.

The figure 1 shows block diagram of 4-bit approximate multiplier circuit using 2-bit multipliers represented by M0, M1, M2 and M3 with 4-bit operands A and B producing partial products i_K, i_L, i_M and i_N of 4-bits each and

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produces the final output o_Y which is of 8-bits using the exact and inexact adders.

The sum and carry equations of exact half adder are:

$$s = a \text{ xor } b \quad (1)$$

$$cy = a.b \quad (2)$$

The sum and carry equations of exact full adder are:

$$sum = A \text{ xor } B \text{ xor } C \quad (3)$$

$$cout = A.B+(Cin.(A \text{ xor } B)) \quad (4)$$

The sum and carry equations of inexact half adder are:

$$s = a+b \quad (5)$$

$$cy = a.b \quad (6)$$

The sum and carry equations of inexact full adder are:

$$sum = !(cout) \quad (7)$$

$$cout = A.B+B.Cin+Cin.A \quad (8)$$

The figure 2 shows the partial product generation and approximation methodology used in 4-bit approximate multiplier architecture. The array representation with two 4-bit operands A and B producing partial products P[15:0] which is of 16-bits from an array of 16 and gates can be seen in this figure. Partial Products P[0] is produced from the and gate array, P[1], P[4], P[2], P[5] and P[8] move to the OR gate array, P[3], P[6], P[7], P[14], P[13] and P[15] move to the exact/inexact adder array P[9], P[10], P[11], and P[12] along with the sum and carry values from the exact/inexact adder array move to the 4-bit adder array and produces the final output o_Y[7:0] of 8-bits.

The figure 3 shows block diagram of 8-bit approximate multiplier circuit using 4-bit multipliers represented by M0, M1, M2 and M3 with 8-bit operands A and B producing partial products i_K, i_L, i_M and i_N of 8-bits each and produces the final output o_Y which is of 16-bits using the exact and inexact adders.

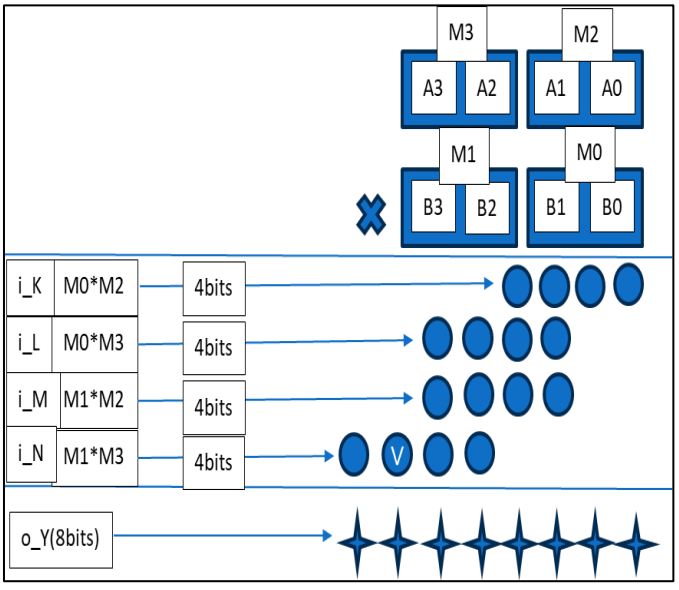


Fig. 1. Partial Products generation and multiplication mechanism of 4-bit Approximate Multiplier

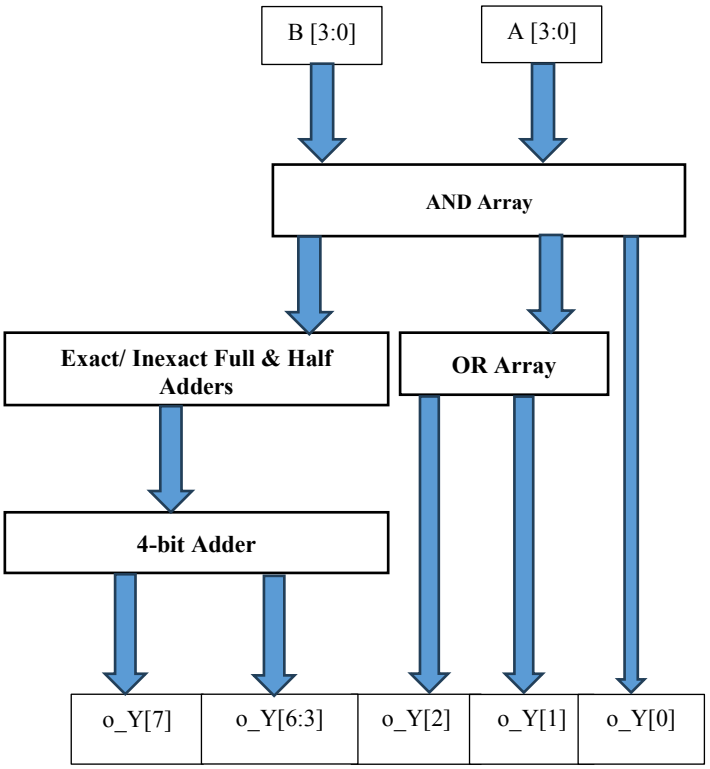


Fig. 2. 4-bit Approximate Multiplier in Array representation

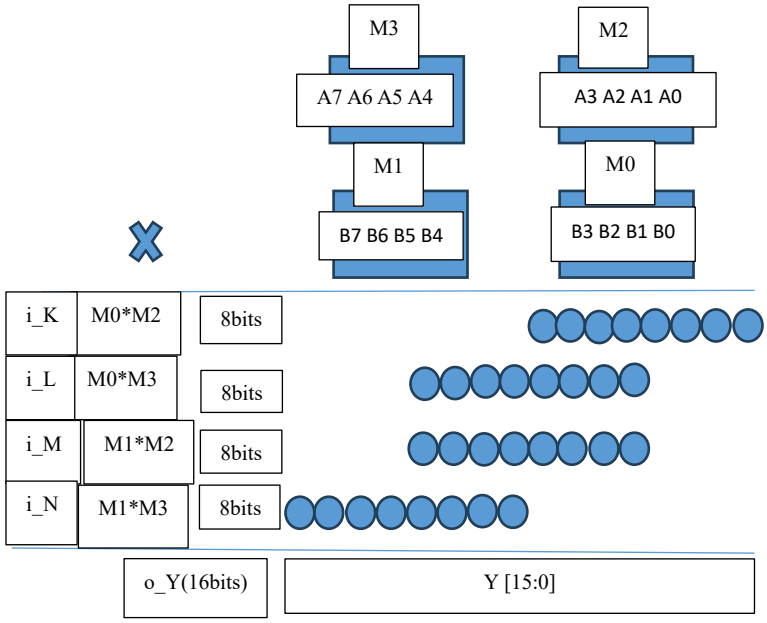


Fig. 3. Block diagram of 8-bit Approximate Multiplier

B. Error Prediction and Correction Logic (EPCL)

Implementing the Error Prediction and Correction Logic (EPCL) for designing approximate multipliers instead of inexact adders have the following advantages:

- The carry chain till the Most Significant Bit (MSB) is eliminated by introducing 3 input OR logic during the product output generation.
- XOR logic is replaced with OR logic which is significant in area (LUT) optimization but error propagation (error factor) across the circuit keeps

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increasing in each stage of the accumulation stage.

- Elimination of the exact/inexact adder stage, specifically for carry propagation through the multiplication process, optimizes power as well as area of the design.
- This logic optimizes delay of the circuit which results in increased performance of the design.

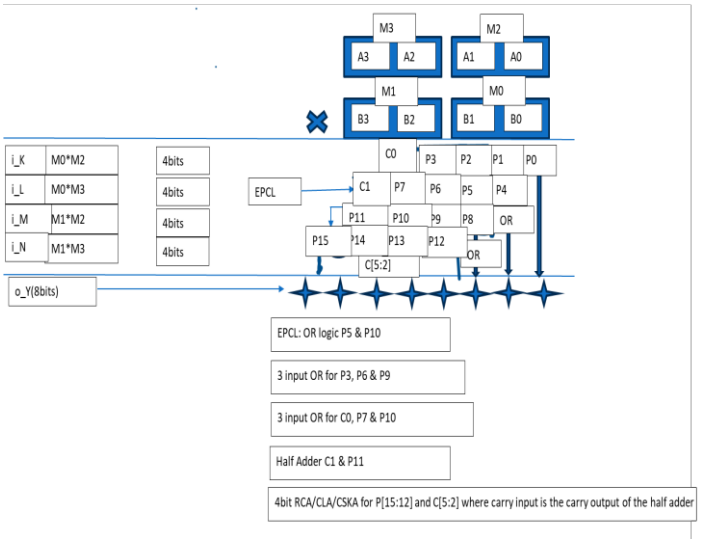


Fig. 4. Error Prediction and Correction Logic for 4-bit approximate multiplier

The EPCL block diagram for 4-bit approximate multiplier circuit using 2-bit multipliers represented by M0, M1, M2 and M3 with 4-bit operands A and B producing partial products i_K, i_L, i_M and i_N of 4-bits each and produces the final output o_Y which is of 8-bits and instead of using exact and inexact adders, the circuit design uses OR logic instead of XOR logic used in exact adder.

III. Methodology

Before There are two steps followed for the proposed work. At first the 4bit approximate multipliers are designed using the inexact adders along with the RCA/CSA/CSKA/CLA adders and EPCL along with RCA/CSKA/CLA adders and also 8bit approximate multiplier using 4bit approximate multiplier. Secondly the two parameters such as power and area utilization for 4bit approximate multiplier are observed after implementation in the VIVADO 2024.2 design tool and tabulated for comparison.

A. Multiplier Design based on different Adders

- Multipliers are of 2 varieties: signed and unsigned.
- Unsigned multipliers use all the bits to express the magnitude of the operand.
- This work focuses on the designs of unsigned multipliers using RCA and CSA and in addition, this work proposes the use of Carry Lookahead Adder (CLA) and Carry Skip Adder (CSKA) using exact and inexact adders.
- The values of power and LUTs obtained are compared from existing work [1] and values from the proposed work are tabulated.

B. Comparison between Adders

The power and area comparison of 5 mostly used adder architectures- Ripple Carry Adder (RCA), Carry Save Adder (CSA), Carry Skip Adder (CSKA) and Carry LookAhead Adder (CLA) are tabulated. In Table I, comparison of values of 4-bit adders between the existing work [6] and the proposed work is tabulated.

C. Design of Error Prediction and Correction

The Error Prediction and Correction Logic is applied in the approximate multiplier circuits to remove the carry chain to the MSB part in the accumulation stage of the multiplication process. It is implemented by replacing XOR logic with OR logic when product output is generated.

D. Error Distance Calculations

The Relative Error Distance (RED) can be determined from the equations mentioned, used to determine the amount of accuracy loss in the final product of the designed approximate multipliers compared to the conventional multipliers.

$$RED = (R_c - R_a)/R_c \quad (9)$$

$$ED = (R_c - R_a) \quad (10)$$

$$MED = \sum(\mathrm{ED})/2^{\mathrm{N}} \quad (11)$$

$$MRED = \sum(\mathrm{RED})/2^{N} \quad (12)$$

$$NED = \mathrm{MED}/Rc_{(max)} \quad (13)$$

where R_c is conventional multiplier output and R_a is the approximate multiplier output. The error metrices formulated above for the proposed work are mentioned in Table V and VI for 4-bit and 8-bit approximate multiplier respectively.

IV. Results and Discussion

A. Simulation results of Approximate Multipliers using Exact and Inexact Adders and EPCL

The simulation environment for the proposed work is VIVADO 2024.2 design tool and the simulation results for 4-bit multiplier using exact adders and RCA/CSA/CSKA/CLA, 4-bit approximate multiplier using inexact adders and RCA/CSA/CSKA/CLA, 4-bit approximate multiplier using EPCL and RCA/CLA/CSKA, 8-bit multiplier using RCA/CSA/CSKA/CLA, 8-bit approximate multiplier using inexact adders and RCA/CSA/CSKA/CLA and 8-bit approximate multiplier using EPCL and RCA/CSKA/CLA are shown from figures 5-10 respectively.

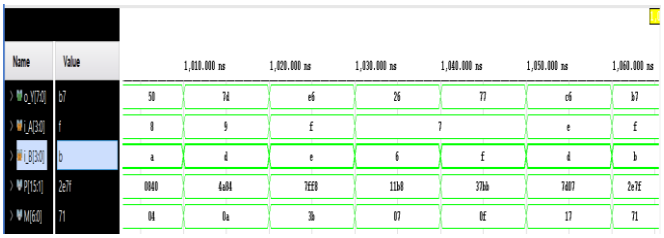


Fig. 5. 4bit multiplier using RCA/CSA/CSKA/CLA

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Fig. 6. 4bit approximate multiplier using inexact adders and RCA/CSA/CSKA/CLA

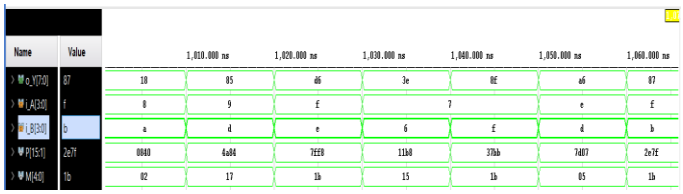


Fig. 7. 4bit approximate multiplier using EPCL and RCA/CLA/CSKA

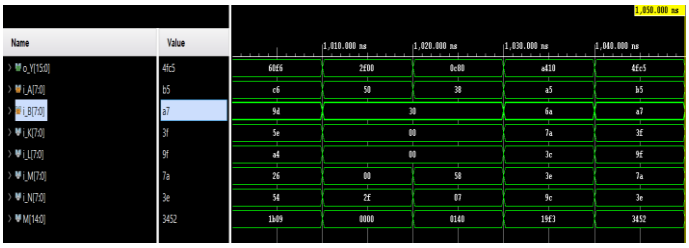


Fig. 8. Simulation of 8bit multiplier using RCA/CSA/CSKA/CLA

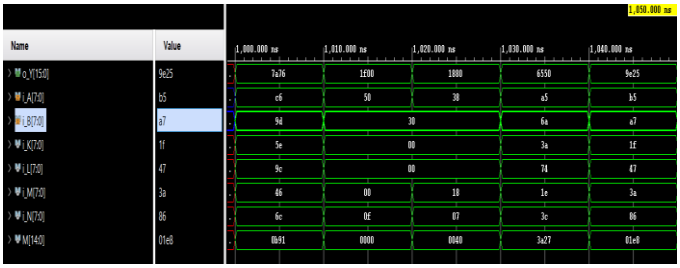


Fig. 9. Simulation of 8bit approximate multiplier using inexact adders and RCA/CSA/CSKA/CLA

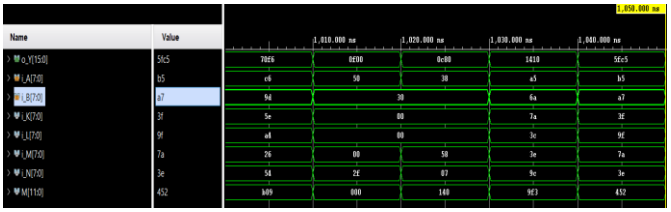


Fig. 10. Simulation of 8bit approximate multiplier using EPCL using RCA/CSKA/CLA

B. *RTL Synthesis Results of Approximate Multipliers using Exact and Inexact Adders and EPCL*

RTL synthesis result of 4-bit approximate multiplier using inexact adders with CSA shown in figure 11 has number of 16 I/O ports, 35 number of nets and 19 number of cells which explains less power and area consumption compared to existing work. The RTL synthesis of 4-bit and 8-bit approximate multiplier using EPCL and RCA/CSKA/CLA are depicted in figures 12 and 13 respectively.

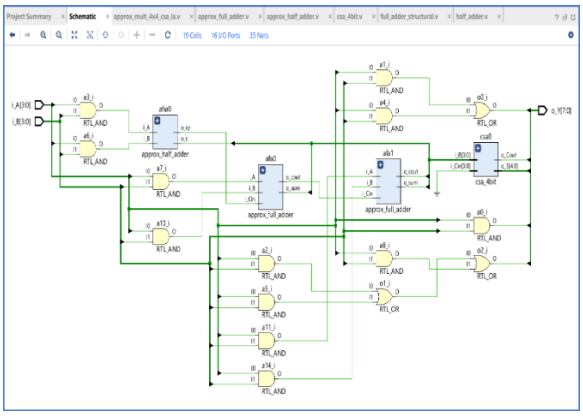


Fig. 11. RTL Circuit of 4bit approx. multiplier using inexact adders and CSA

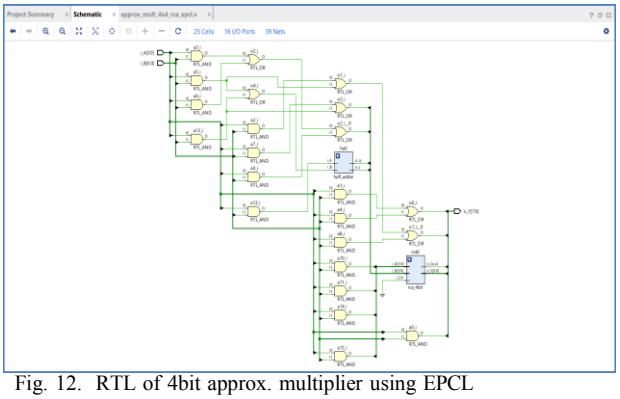


Fig. 12. RTL of 4bit approx. multiplier using EPCL

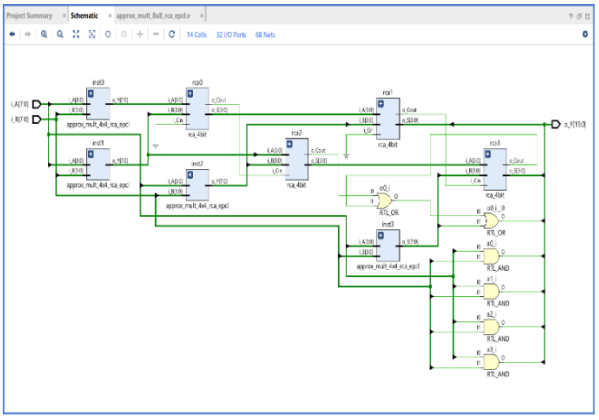


Fig. 13. RTL of 8bit approximate multiplier using EPCL

C. *Comparison between Adders*

TABLE I. COMPARISON OF AREA AND POWER OF 4-BIT ADDERS

ADDERS	POWER (in W) from existing work [5]	POWER (in W) from this work	LUTs from existing work [5]	LUTs from this work
RCA	2.923	2.866	4	4
CSA	-	2.775	-	5
CSKA	2.925	2.865	4	4
CLA	2.948	2.865	4	4

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TABLE II. COMPARISON OF POWER AND AREA OF 4-BIT MULTIPLIER USING RCA/CSA/CSKA/CLA

ADDERS	POWER (in W)	AREA (LUTs)
RCA	0.210	12
CSA	0.169	5
CSKA	0.160	5
CLA	0.160	5

For 4-bit unsigned multiplier using exact adders and RCA/CSKA/CLA/CSA, the power consumption and LUTs are tabulated in Table II.

TABLE III. COMPARISON OF POWER AND AREA OF 4-BIT APPROXIMATE MULTIPLIER USING INEXACT ADDERS

ADDERS	POWER (in W) from existing work [1]	POWER (in W) from this work	LUTs from existing work [1]	LUTs from this work
RCA	0.141	0.141	11	5
CSA	0.142	0.136	11	5
CSKA	-	0.141	-	5
CLA	-	0.141	-	5

For 4-bit approximate multiplier using inexact adders and CSA, the power consumption and LUTs show 4.22% and 54.54% change less shown in Table III.

TABLE IV. COMPARISON OF POWER AND AREA OF 4-BIT APPROXIMATE MULTIPLIER USING EPCL

ADDERS	POWER (in W)	AREA (LUTs)
RCA	0.210	12
CSKA	0.216	8
CLA	0.216	8

Table IV shows the power consumption and LUTs for 4-bit unsigned multiplier using EPCL and RCA/CSKA/CLA.

TABLE V. COMPARISON OF RED FOR 4-BIT APPROXIMATE MULTIPLIER

OPERAND 1(A)	8	9	F	7	7	e	f
OPERAND 2(B)	A	D	E	6	F	d	b
Exact Value after operation	$(50)_{16}$ $(80)_{10}$	$(7d)_{16}$ $(125)_{10}$	$(e6)_{16}$ $(230)_{10}$	$(26)_{16}$ $(38)_{10}$	$(77)_{16}$ $(119)_{10}$	$(c6)_{16}$ $(198)_{10}$	$(b7)_{16}$ $(183)_{10}$
Approximate value after operation	$(30)_{16}$ $(48)_{10}$	$(9d)_{16}$ $(157)_{10}$	$(c6)_{16}$ $(198)_{10}$	$(56)_{16}$ $(86)_{10}$	$(97)_{16}$ $(151)_{10}$	$(96)_{16}$ $(150)_{10}$	$(8f)_{16}$ $(143)_{10}$
Approximate value using EPCL after operation	$(18)_{16}$ $(24)_{10}$	$(85)_{16}$ $(133)_{10}$	$(d6)_{16}$ $(214)_{10}$	$(3e)_{16}$ $(62)_{10}$	$(8f)_{16}$ $(143)_{10}$	$(a6)_{16}$ $(166)_{10}$	$(87)_{16}$ $(135)_{10}$
RED using Inexact Adders with RCA/CSA/CSKA/CLA in %	40	25.6	13.91	126.31	26.89	24.24	21.86
RED using EPCL with RCA, CSKA, CLA in %	70	6.4	6.9	63.16	20.17	16.16	26.23

Table V shows the relative error distance(RED) tabulation and the Mean RED(MRED), Mean Error Distance(MED) and Normalized RED(NRED) for 4-bit unsigned multiplier are 2.18%, 2.06 and 0.89 respectively using inexact adders and 1.63%, 1.625 and 0.71 respectively using EPCL.

TABLE VI. COMPARISON OF RED FOR 8-BIT APPROXIMATE MULTIPLIER

OPERAND 1(A)	c6	50	38	a5	b5
OPERAND 2(B)	9d	30	30	6a	a7
Exact Value after operation	$(60f6)_{16}$ $(24822)_{10}$	$(2f00)_{16}$ $(12032)_{10}$	$(0c80)_{16}$ $(3200)_{10}$	$(a410)_{16}$ $(42000)_{10}$	$(4fc5)_{16}$ $(20421)_{10}$
Approximate value after operation	$(7a76)_{16}$ $(31350)_{10}$	$(1f00)_{16}$ $(7976)_{10}$	$(1880)_{16}$ $(6272)_{10}$	$(6550)_{16}$ $(26000)_{10}$	$(9e25)_{16}$ $(39973)_{10}$
Approximate value using EPCL after operation	$(70f6)_{16}$ $(28918)_{10}$	$(0f00)_{16}$ $(3840)_{10}$	$(0c80)_{16}$ $(3200)_{10}$	$(1410)_{16}$ $(5136)_{10}$	$(5c5f)_{16}$ $(24517)_{10}$
RED using Inexact Adders with RCA/CSA/CSKA /CLA in %	26.29	34.04	96	38.09	95.74
RED using EPCL with RCA, CSKA, CLA (%)	16.50	68.08	-	87.77	20.06

Table VI shows the relative error distance(RED) tabulation and the Mean RED(MRED), Mean Error Distance(MED) and Normalized RED(NRED) for 8-bit unsigned multiplier are 9.07%, 1537.75 and 0.036 respectively using inexact adders and 6.01%, 1664 and 0.039 respectively using EPCL and value of RED after approximate multiplication using EPCL for operands $(38)_{16}$ and $(30)_{16}$ resulted in no change in the exact and approximate product.

CONCLUSION

This work focuses on designing low power and area (LUTs) consuming unsigned 4-bit and 8-bit approximate multipliers using exact and inexact adders with RCA/CLA/CSKA/CSA adders and using EPCL with RCA/CLA/CSKA. The results for 4-bit approximate multiplier using inexact and CSA, the power consumption and LUTs show 4.22% and 54.54% change less compared to the existing work. Additionally, the proposed 4-bit unsigned approximate multiplier using inexact adders with CSKA and CLA evaluates power consumption to 0.141 W and LUTs to 5, like the results of 4-bit unsigned multiplier using inexact adder and RCA shown in Table 2. From this work we found that approximate multiplier design using inexact adders are more power and area optimized rather than approximate multiplier design using exact adders and EPCL as evident from Table II, III and IV. Hence, approximate multiplier using inexact adders would be more beneficial for designing processing elements(PE) in AI accelerators in future work.

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